

Xinyang Geng

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EDUCATION

University of California, Berkeley

Ph.D. in Computer Science

Aug 2018 – Now

- Advised by Professor Sergey Levine.
- Research in large language models, vision-language models and deep reinforcement learning.

University of California, Berkeley

B.A. in Computer Science and Statistics (double major)

Aug 2013 – May 2017

- Graduated with high distinction and honor in computer science.

PUBLICATIONS AND PREPRINTS

Offline Q-Learning on Diverse Multi-Task Data Both Scales And Generalizes

Aviral Kumar*, Rishabh Agarwal*, **Xinyang Geng**, George Tucker, Sergey Levine
Notable top 5%, ICLR 2023, arXiv:2211.15144

Towards Better Few-Shot and Finetuning Performance with Forgetful Causal Language Models

Hao Liu*, **Xinyang Geng***, Lisa Lee, Igor Mordatch, Sergey Levine, Sharan Narang, Pieter Abbeel
arXiv:2210.13432

Multimodal Masked Autoencoders Learn Transferable Representations

Xinyang Geng*, Hao Liu*, Lisa Lee, Dale Schuurams, Sergey Levine, Pieter Abbeel
arXiv:2205.14204

Design-bench: Benchmarks for data-driven offline model-based optimization

Brandon Trabucco*, **Xinyang Geng***, Aviral Kumar, Sergey Levine
Oral presentation, ICML 2022, arXiv:2202.08450

Conservative Objective Models for Effective Offline Model-Based Optimization

Brandon Trabucco*, Aviral Kumar*, **Xinyang Geng**, Sergey Levine
Oral presentation, ICML 2021, arXiv:2107.06882

Variable-Shot Adaptation for Online Meta-Learning

Tianhe Yu*, **Xinyang Geng***, Chelsea Finn, Sergey Levine
arXiv:2012.07769

Meta-Reinforcement Learning Robust to Distributional Shift via Model Identification and Experience Relabeling

Russell Mendonca*, **Xinyang Geng***, Chelsea Finn, Sergey Levine
arXiv:2006.07178

Rewriting History with Inverse RL: Hindsight Inference for Policy Improvement

Benjamin Eysenbach*, **Xinyang Geng***, Sergey Levine, Ruslan Salakhutdinov
Oral presentation, NeurIPS 2020, arXiv:2002.11089

Dynamical Distance Learning for Unsupervised and Semi-Supervised Skill Discovery

Kristian Hartikainen, **Xinyang Geng**, Tuomas Haarnoja*, Sergey Levine*
ICLR 2020. arXiv:1907.08225

Improved Generalization with Curvature Regularization

Xinyang Geng, Lechao Xiao, Hossein Mobahi, Jeffrey Pennington
ICML 2018 Workshop: Modern Trends in Nonconvex Optimization for Machine Learning Workshop

Automatic Goal Generation for Reinforcement Learning Agents

Carlos Florensa*, David Held*, **Xinyang Geng***, Pieter Abbeel
ICML 2018. arXiv:1705.06366

Real-Time User-Guided Image Colorization with Learned Deep Priors

Richard Zhang*, Jun-Yan Zhu*, Phillip Isola, **Xinyang Geng**, Angela S. Lin, Tianhe Yu, Alexei A. Efros
SIGGRAPH 2017. arXiv:1705.02999

Deep Reinforcement Learning for Tensegrity Robot Locomotion

Marvin Zhang*, **Xinyang Geng***, Jonathan Bruce*, Ken Caluwaerts, Massimo Vespignani, Vytas SunSpiral, Pieter Abbeel, Sergey Levine
ICRA 2017. arXiv:1609.09049

**RESEARCH
EXPERIENCE**

UC Berkeley

Graduate Student Researcher

Aug 2018 – Now
Berkeley, CA

- Advised by Professor Sergey Levine.
- Working on deep reinforcement learning, model-based optimization, and language models.

Google Brain

Research Intern

May 2022 – October 2022
Mountain View, CA

- Worked under the supervision of Igor Mordatch
- Worked on research projects in large language models, vision-language models and deep reinforcement learning.

Google Brain

Google AI Resident

Jul 2017 – Jul 2018
Mountain View, CA

- Worked under the supervision of Sergey Levine, Jeffrey Pennington, Hossein Mobahi, Nicolas Le Roux and Ofir Nachum.
- Worked on research projects in neural network architecture improvement, regularization techniques for neural networks and meta-reinforcement learning.

UC Berkeley AI Research Lab

Undergraduate Research Assistant

Oct 2015 – May 2017
Berkeley, CA

- Worked under the supervision of Professor Pieter Abbeel, Professor Sergey Levine and Professor Alexei Efros.
- Performed research on deep reinforcement learning for robotics, curriculum for goal conditioned reinforcement learning and image colorization.

ENGINEERING EXPERIENCE

Google

Software Engineering Intern

May – Aug 2015
Mountain View, CA

- Worked in advertisement data infrastructure engineering productivity team.
- Designed and built dashboard website for a log processing pipeline testing framework.
- Created performance visualization and alert service.

IBM

Software Engineering Intern

May – Aug 2014
Emeryville, CA

- Worked on the IBM Endpoint Manager product.
- Designed software for multiple platforms including Windows with Microsoft SQL Server and Linux with IBM DB2.
- Developed GUI in Microsoft Visual Studio with MFC.

RESEARCH SOFTWARE

Koala: A Dialogue Model for Academic Research

Blog post: <https://bair.berkeley.edu/blog/2023/04/03/koala>

Online demo: <https://koala.lmsys.org/>

Fine-tuned on top of Meta’s LLaMA, Koala is a dialogue language model trained on dialogue data gathered from the web. Our user study shows that Koala can effectively respond to a variety of user queries, generating responses that are often preferred over Alpaca, and at least tied with ChatGPT in over half of the cases.

EasyLM

GitHub: <https://github.com/young-geng/EasyLM>

Large language models (LLMs) made easy, EasyLM is a one stop solution for pre-training, fine-tuning, evaluating and serving LLMs in JAX/Flax. EasyLM can scale up LLM training to hundreds of TPU/GPU accelerators by leveraging JAX’s pjit compilation.

MLXU

GitHub: <https://github.com/young-geng/mlxu>

Machine Learning eXperiment Utilities: convenient utilities for running machine learning experiments, parsing experiment configurations and logging results.

JaxCQL

GitHub: <https://github.com/young-geng/JaxCQL>

A simple and modular implementation of the conservative Q learning (CQL) and soft actor critic (SAC) algorithm in JAX and Flax. This implementation is highly efficient, so that you can train an SAC agent in 30 minutes and a CQL agent in an hour.

RESEARCH MENTORING

Russell Mendonca (2019-2020)

Undergraduate at UC Berkeley → PhD student at CMU.
CRA Undergraduate Research Award finalists.

Brandon Trabucco (2020-2021)

Undergraduate at UC Berkeley → PhD student at CMU.

Kevin Li (2022)

5th year MS at UC Berkeley.

Sathvik Kolli (2022-2023)

5th year MS at UC Berkeley, to graduate in 2023.

Han Qi (2022-2023)

Undergraduate at UC Berkeley, to graduate in 2023.

CRA Undergraduate Research Award nomination.

SERVICES

Reviewer: ICML 2019, NeurIPS 2019, ICLR 2020, NeurIPS 2020, ICLR 2021, ICML 2021, NeurIPS 2021, Neurips 2022, ICLR 2022, ICLR 2023

**TECHNICAL
SKILLS**

Deep Learning Frameworks: JAX, PyTorch

Github Profile: <https://github.com/young-geng>